Yoganandh Madhuranthakam

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EDUCATION

PhD in Electrical and Computer Engineering Jan 2021 – Aug 2025
Michigan State University, East Lansing

- Dissertation: Nonlinear acoustic wave propagation in inhomogeneous media.
- Supervisor: Dr. Sunil Kishore Chakrapani | CGPA: 3.75/4.0

Masters in Mechanical Engineering Aug 2016 – Jun 2018
Indian Institute of Technology (IIT) Hyderabad, Hyderabad

- Dissertation: Prediction of breakout noise from a rectangular duct using SEA and deterministic methods.
- Supervisor: Dr. Venkatesham B| CGPA: 9.71/10

RESEARCH EXPERIENCE

Graduate Research Assistant, **Michigan State University** – East Lansing, MI Jan 2021 – Present

- Developed a novel inversion technique to measure the absolute acoustic nonlinearity parameter β using a **nonlinear immersion ultrasonic test** and a theoretical model of nonlinear acoustic wave propagation
- Established a theoretical model to capture the effect of internal stress and dislocations on β in the **nonlinear resonant ultrasound spectroscopy(NRUS)** study of cantilever beam
- Developed a theoretical framework for capturing the diffraction of a **nonlinear acoustic wave** propagating through *n*-layer media and validated it with **numerical simulations (COMSOL and k-wave)**
- Performed characterization of surface cracks on a railroad through **ultrasonic flaw detection experiments** and **finite element study** of Rayleigh wave propagation
- Designed and performed a material characterization study on ultrasonically welded titanium and copper samples using the **nonlinear ultrasonic** and **shear birefringence** tests

Graduate Research Assistant, **Indian Institute of Technology (IIT)** – Hyderabad Aug 2016 – Jun 2018

- Implemented a new emerging acoustic analysis technique called **Statistical Energy Analysis (SEA)** in predicting breakout noise from an HVAC duct
- Performed a comparison study between SEA and existing **numerical techniques (FEM/BEM) in COMSOL**

INDUSTRIAL EXPERIENCE

NVH Engineer, W.M. Associates Inc, Burton, Ohio Oct 2025 – Present

- Development of NVH test software to predict nicked gears in Ford EV axle.
- Support OEM and supplier NVH programs through experimental testing and advanced vibration analysis, delivering diagnostic insights via custom signal-processing and visualization tools.

System Modelling Engineer, **Eaton India Innovation Centre, Eaton Corporation**, Aug 2018 – Dec 2020
Pune, India

- Designed and developed a **digital prototype** of a heavy-duty commercial vehicle to conduct **dynamic analysis**, evaluating the impact of clutch, transmission, and differential on **torsional vibrations**
- Developed a high-fidelity dynamic model of a commercial vehicle to simulate **pass-by noise tests (SAE J366, ISO 362)** with an automated process using a MATLAB-based GUI
- Optimized the design of the Inertia Brake for heavy-duty vehicle transmissions, defining critical quality parameters through **mathematical/ physical models and FEA simulations**. Earned **DFSS Green Belt certification**
- Introduced an **SEA**-based approach for transmission sound radiation analysis, reducing computation time by three weeks per case and saving $ 10,000 annually compared to **FEM/BEM** methods

AWARDS and ACHIEVEMENTS

Fellowship Award, ASNT (American Society for Nondestructive Testing) 2023
Travel Grant Award, Michigan State University College of Engineering 2022, 2023 and 2024
Green Belt in Design for Six Sigma Methods (DFSS), from Eaton Corporation for performing design optimization of Eaton's automotive product - Inertia Brake 2019
Teaching Assistantship Award, Ministry of Human Resource Development India 2016

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant, **Michigan State University** – East Lansing, MI Sep 2024 – Dec 2024

- **Course**: ECE 415 Computer Aided Manufacturing
  Description : Tutored a class of 20 students in CAD/CAM Laboratory, covering milling and lathe machines, programmable controllers, numerical control, NC part programming, sensors, and data acquisition systems

Teaching Assistant, **Michigan State University** – East Lansing, MI Jan 2023 – Apr 2023

- **Course**: ECE 416 Digital Control
  Description : Tutored a class of 36 students in Digital Control Laboratory, covering state-space modeling, analysis, and design of control systems, with simulations of nonlinear systems such as suspended double pendulum, thermostats, and stepper motors

Mentor, **PUMA/NDEL labs, Michigan State University** – East Lansing, MI Jan 2023 – Dec 2024

- **Description** : Mentored two PhD and two Master's students in numerical modeling of elastic and acoustic waves using ANSYS and COMSOL, and in experimental techniques for ultrasonic Nondestructive Evaluation

Publications

Peer-reviewed:

1. **Madhuranthakam Y**, Chakrapani SK. Diffraction and Attenuation of Nonlinear Acoustic Waves in Layered Media. Journal of Theoretical and Computational Acoustics. 2026 34:02. https://doi.org/10.1142/S2591728526500015.
2. **Madhuranthakam, Y.**, Barnard, D. J., & Chakrapani, S. K. (2025). Inversion of Acoustic Nonlinearity Parameter of a Solid Using Immersion Method. Research in Nondestructive Evaluation, 36(3), 105–123. https://doi.org/10.1080/09349847.2025.2489932
3. **Madhurnathakam, Y.**, & Chakrapani, S. (2026). A simple reference-based method for measuring acoustic nonlinearity parameter of solids using immersion ultrasonics. Frontiers in Nondestructive Testing and Evaluation: Academic Insights. Research and Review Journal of Nondestructive Testing Vol. 4. https://doi.org/10.58286/32881
4. **Madhuranthakam Y**, Chakrapani SK. Analytical Study of Nonlinear Flexural Vibration of a Beam with Geometric, Material and Combined Nonlinearities. Vibration. 2024; 7(2):464-478. https://doi.org/10.3390/vibration7020025.
5. **Madhuranthakam Y**, Nagaraja Jade and Venkatesham B. "Prediction of Insertion loss of lagging in rectangular duct using Statistical Energy Analysis", Noise Control Engineering Journal, Volume 67,1 November 2019, pp. 438-446(9). https://doi.org/10.3397/1/376740.
6. **Madhuranthakam Y**, Nagaraja Jade and Venkatesham B. "A simplified model to predict the transverse transmission loss using radiation efficiency", The Journal of Acoustical Society of India, Volume 46, Number 3-4, December 2019, pp. 145-150.
7. Sharma, S., **Madhuranthakam, Y.**, Vu, A., Khasawneh, F. A., Poudel, A., Deng, Y., & Chakrapani, Sunil. K. (2026). Characterisation of rolling contact fatigue cracks using Rayleigh wave scatter field aided by machine learning. Nondestructive Testing and Evaluation, 1–20. https://doi.org/10.1080/10589759.2026.2669840
8. A. Vu, **Madhuranthakam Y**, A. Poudel, S. K. Chakrapani, Numerical study of Rayleigh wave interaction with rolling contact fatigue type of defects, Research in Nondestructive Evaluation, Feb 02-07 2023, doi: 10.1080/09349847.2023.2180560.
9. A. Vu, **Madhuranthakam Y**, S. K. Chakrapani, Detection and Characterization of Rolling Contact Fatigue Type of Defects Using Rayleigh Surface Waves, Proceedings of ASNT research conference, St. Louis, June 06-20 2022, doi: 10.32548/RS.2022.019.

Articles under preparation and review:

1. **Madhuranthakam Y**, Syed Akbar Ali M, Barnard D, Chakrapani SK, A study on the effect of internal stress on the acoustic non-linearity parameter using non-linear ultrasound spectroscopy (NRUS), (Under preparation).
2. **Madhuranthakam Y**, Chakrapani SK. Evaluation of residual stress in acoustoplastically deformed copper structures using shear birefringence (Under preparation).

Conference and poster presentations:

1. **Madhuranthakam Y**, Chakrapani SK, Nonlinear waves in layered media, Presentation in 49th Annual Quantitative Nondestructive Evaluation (QNDE) conference, San Diego, July 25-27, 2022.
2. **Madhuranthakam Y**, Syed Akbar Ali M, Barnard D, Chakrapani SK, A study on the effect of internal stress on the acoustic non-linearity parameter using non-linear ultrasound spectroscopy, 32nd American Society for Nondestructive Testing Research Symposium, Pittsburgh, June 24-28, 2024.
3. **Madhuranthakam Y** , Barnard D , Chakrapani SK, Inversion of acoustic nonlinearity parameter of solid using immersion method, 32nd American Society for Nondestructive Testing Research Symposium, Pittsburgh, June 24-28, 2024.
4. **Madhuranthakam Y**, Chakrapani SK. "Nonlinear waves in layered media ". Graduate Research Symposium, Michigan State University, East Lansing, May 2023 (Poster).
5. **Madhuranthakam Y**, Barnard D , Chakrapani SK. "Inversion of acoustic nonlinearity parameter of solid using immersion method". Graduate Research Symposium, Michigan State University, East Lansing, May 2024 (Poster).

TECHNOLOGIES/SKILLS

- **Scripting:** MATLAB, C, C++, Python and LaTex
- **Computational and computer-aided packages:** ANSYS (APDL and Workbench), Comsol, K-wave, Simulink, Mathematica, AMESim, GT - Suite, ESI VA one, and CAD(Solid works and Fusion 360)
- **Experimental:** Contact and immersion transducers, Acoustic imaging (A-scan, B-scan, and C-scan), Signal generator, Gated RF pulse amplifier, Oscilloscope (12 bit, 200 MHz), Transducer calibration
- **Soft skills:** Problem-solving, Independent research documentation, Time Management, Teamwork
- **Languages:** English, Telugu, Hindi and Tamil.